

COMP1117 Tutorial 1

INTRODUCTION · PYTHON + VS CODE

About the tutorials

- ▶ Tutorial 1: Introduction
 - ▶ Course arrangement reminder: pre-class video checkpoints
 - ▶ What an editor is, what Python is, and why you need both
 - ▶ Installation time — install Python + VS Code
 - ▶ Your first program: write hello.py and run it
 - ▶ Two practice tasks, if everything is working

Pre-class video checkpoints (10%)

- ▶ Each week's videos are followed by a short **multiple-choice checkpoint**, to be completed **before that week's lecture starts**. Class time assumes you have already met the material.
- ▶ The **first** checkpoint does **not** count towards your mark, because it falls inside the add-drop period.
- ▶ A checkpoint you do not complete **scores zero**, and **none are dropped**.

Tutorial arrangement

- ▶ We write our programs in **VS Code** on your own computer. Today we set it up together.
- ▶ Bring your **laptop** to every tutorial.
- ▶ If you have a question, ask the tutor during the tutorial or post on the Moodle Forum.

Today's plan

- ▶ **Part 1** • What are Python and VS Code, and why do we need both?
- ▶ **Part 2 • Installation time.** Everyone installs Python, VS Code and the Python extension, following the slides step by step.
 - ▶ **Already installed at home?** Run the check on the “Check: is Python installed?” slide, then help your neighbour or start Part 3.
- ▶ **Part 3** • Your first program: create hello.py and run it — in a terminal, and with the ▶ button.
- ▶ **Part 4** • Two practice tasks for everyone whose Hello World works.

Goal for today: everyone leaves with Python + VS Code working and a bug-free Hello World running.

What happens when you “run a program”?

You write text

```
print("Hello World")
```

a plain text file, e.g. hello.py



Python reads it

The Python interpreter, a program installed on your computer, executes it line by line



Result appears

```
Hello World
```

on the screen

Two different things are involved: the place where you WRITE the text, and the program that RUNS it.

Two things you need

1. Python (the “translator”)

- ▶ A program that understands Python code and executes it
- ▶ Without it, your computer does not “speak Python”
- ▶ Works in the background — it has no nice window to type in
- ▶ Free, from **python.org**

2. VS Code (the “notebook”)

- ▶ An editor: a place to type code, save it as a file, and press one button to ask Python to run it
- ▶ Colours your code, points out typos, shows the output
- ▶ By itself it CANNOT run Python — it calls the translator
- ▶ Free, from **code.visualstudio.com**

Editor vs Python — common confusions

“I installed VS Code, why can’t I run my program?”	You have the notebook but no translator. Install Python.
“I installed Python, why is there no window to type in?”	Python works in the background. You need an editor (VS Code) to write in.
“Can I write my program in Word / Notes / WeChat?”	You can type the text, but it is not saved as a plain .py file and there is no run button. Use VS Code.
“Do I install them every time?”	No. Install Python, VS Code and the Python extension ONCE. After that, just open VS Code.

Where does a program live?

- ▶ A Python program is a **plain text file** whose name ends with **.py**, e.g. `hello.py`
- ▶ It is saved in a folder on your computer, like any other file
- ▶ You can save it in any folder you like — just remember where it is
- ▶ In today's examples we use a folder called **COMP1117** on the Desktop
 - ▶ VS Code opens one folder at a time, and its terminal starts inside that folder. (A terminal is another way to control your computer: instead of clicking icons and menus, you type an instruction — a command — and the computer carries it out and answers in text. That is how we will tell Python to run a file.)

Desktop / COMP1117 (example)

```
hello.py  
ask.py  
ex1_1.py  
ex1_2.py  
...
```

Step 1 — Install Python (Windows)

- ▶ Open <https://www.python.org/downloads/>
- ▶ Click the yellow button **Download Python 3.14.x** and open the downloaded file
- ▶ On the first screen, **TICK** ☐ **Add python.exe to PATH** (bottom of the window)
- ▶ Click **Install Now**, wait, then **Close**

Why tick this box? PATH is the list of folders Windows searches when you type a command. Ticking the box adds Python to that list. Later, when you type python in a terminal — or VS Code types it for you — Windows knows where the translator is.

Not ticked → “python' is not recognized...” or the Microsoft Store opens, and nothing runs.

Install Python 3.14.4 (64-bit)

→ **Install Now**

Customize installation

☒ Use admin privileges when installing py.exe

☒ **Add python.exe to PATH**

(what the installer looks like)

Step 1 — Install Python (Mac)

- ▶ Open <https://www.python.org/downloads/>
- ▶ Click the yellow button **Download Python 3.14.x** — you get a file ending with .pkg
- ▶ Double-click it, then **Continue → Continue → Agree → Install**
- ▶ Type your Mac login password when asked, then Close

Check: is Python installed?

Windows

- ▶ Press the **Windows key**, type **cmd**, press Enter
- ▶ A black window (Command Prompt) opens
- ▶ Type **python --version** and press Enter

You should see something like this (the last digits may differ):

```
Python 3.14.4
```

Mac

- ▶ Press **Cmd + Space**, type **Terminal**, press Enter
- ▶ A white or black window (Terminal) opens
- ▶ Type **python3 --version** and press Enter

The ONLY platform difference to remember: Windows types `python` • Mac types `python3`

What is a terminal? Another way to control your computer: instead of clicking icons and menus, you type an instruction (a command), press Enter, and the computer carries it out and answers in text. Windows calls it Command Prompt, Mac calls it Terminal — same idea.

If it doesn't work...

Windows

- ▶ You see '**python**' is not recognized... or the **Microsoft Store** opens
- ▶ → the PATH box was not ticked. Run the installer again → **Modify** → Next → tick **Add Python to environment variables** → Install
- ▶ Then **restart the computer** and try again

Mac

- ▶ A pop-up says “**install the command line developer tools?**”
- ▶ → click Install and wait, or cancel and use the python.org installer (previous slide)
- ▶ You see **command not found: python3**
- ▶ → the python.org installer was not run, or the Terminal was opened before installing. Close Terminal and open it again.

Step 2 — Install VS Code

- ▶ Open <https://code.visualstudio.com/> and click the big blue **Download** button

Windows

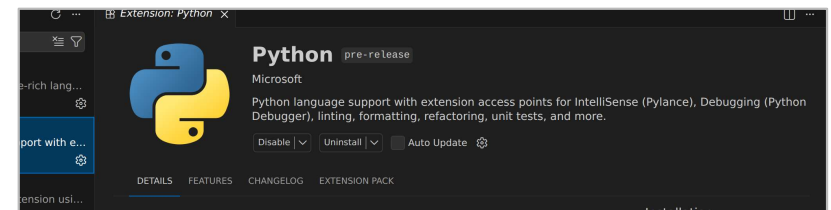
- ▶ Open the downloaded VSCodeUserSetup-... .exe
- ▶ Accept the agreement → Next → Next ... → Install
- ▶ Tip: keep “**Add to PATH**” ticked (it is by default)
- ▶ Finish → VS Code opens

Mac

- ▶ Double-click the downloaded .zip to unpack it
- ▶ Drag **Visual Studio Code** into the **Applications** folder
- ▶ Open it from Applications (or Launchpad). If asked “**downloaded from the Internet, open?**” → Open

Step 3 — Install the Python extension

- ▶ In VS Code, click the **Extensions** icon on the left bar, or press Ctrl+Shift+X / Cmd+Shift+X
- ▶ Type **Python** in the search box
- ▶ Choose the one named **Python** published by **Microsoft** → click **Install**
- ▶ It also installs Pylance and Python Debugger for you — that is normal
- ▶ **Do NOT install “Code Runner”** — it cannot handle programs that ask for input



You will see an Install button here instead of Disable/Uninstall

Do I have to redo this every time?

- ▶ **No.** Python, VS Code and the Python extension are installed **once**.
- ▶ From now on, every time you want to program:
 - ▶ open VS Code
 - ▶ open the folder with your programs (it usually reopens by itself)
 - ▶ write, run, save
- ▶ If you switch to a new computer, repeat Steps 1–3 on that computer.

Step 4 — Make a folder for today

Windows

- ▶ Right-click an empty spot on the **Desktop**
- ▶ **New → Folder**
- ▶ Name it **COMP1117** and press Enter

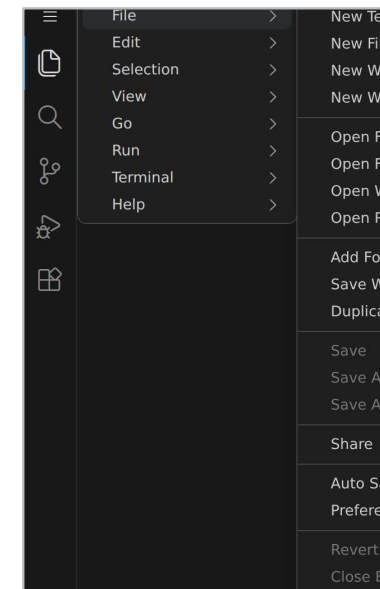
Mac

- ▶ Right-click an empty spot on the **Desktop**
- ▶ **New Folder**
- ▶ Name it **COMP1117** and press Return

- ▶ Why a folder? VS Code works with one folder at a time — EXPLORER shows the files in that folder, and the terminal starts inside it.
- ▶ Why does that matter? When you type `python hello.py`, Python looks for `hello.py` in the folder the terminal is in. Same folder → it runs.
- ▶ Different folder → “can't open file 'hello.py': No such file or directory”. Opening the right folder in VS Code avoids this.
- ▶ Any folder and any name would work. We all use Desktop / COMP1117 today so that the slides match your screen — later you can organise your files however you like

Open the folder in VS Code

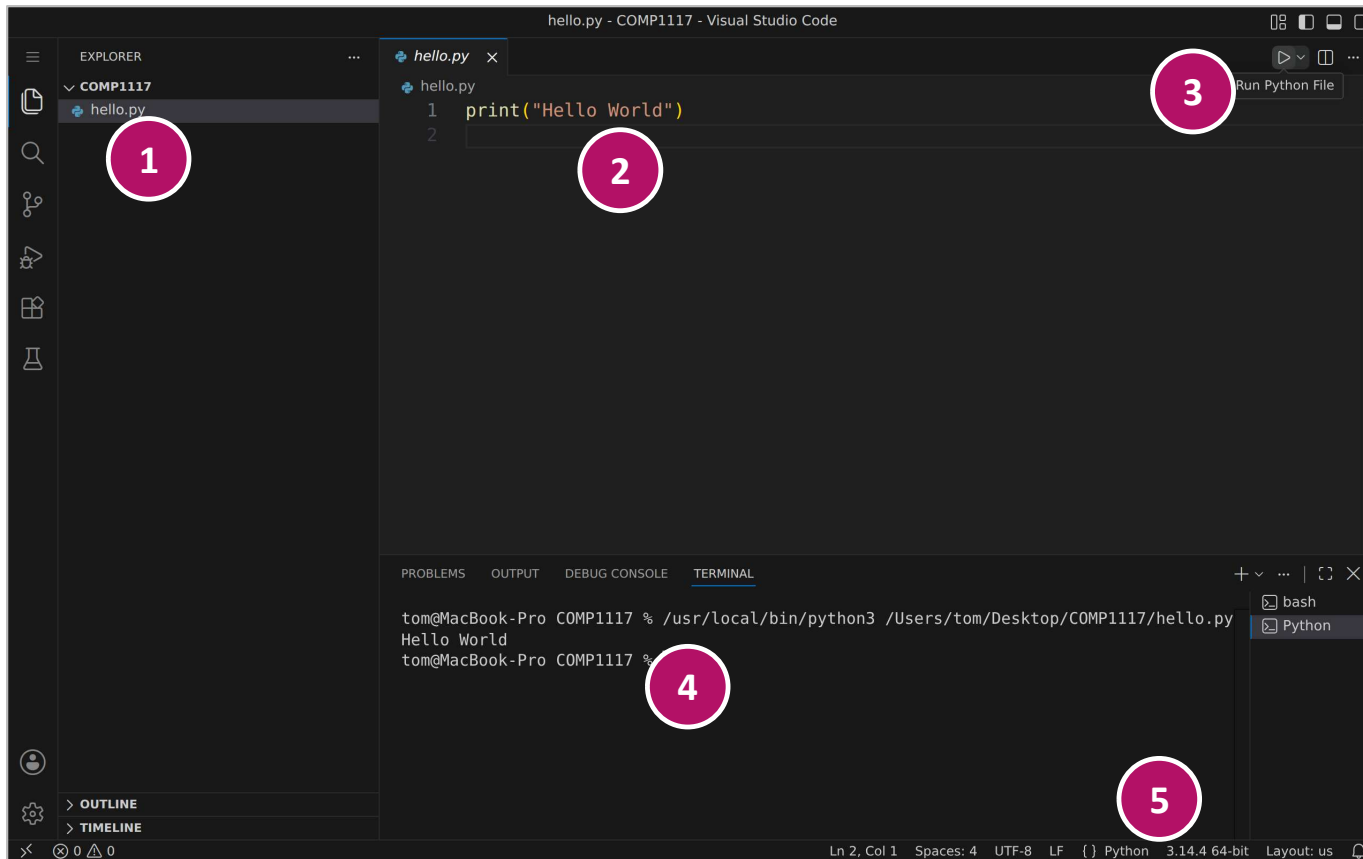
- ▶ In VS Code: menu **File** → **Open Folder...** (Mac: **File** → **Open Folder...** in the top menu bar)
- ▶ Choose **Desktop** → **COMP1117** → click **Select Folder / Open**
- ▶ A box asks “Do you trust the authors...?” → **Yes, I trust the authors**
- ▶ The left panel (**EXPLORER**) now shows **COMP1117** — empty for now



Open Folder...

the File menu

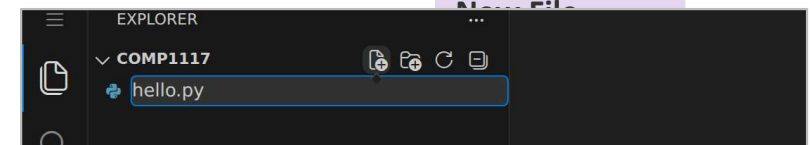
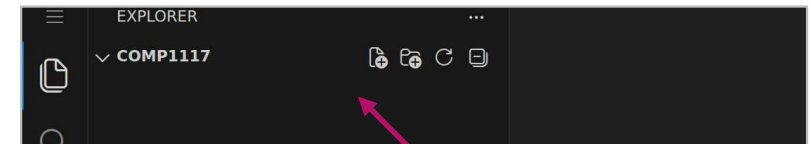
Tour of the window



- 1 EXPLORER — your files
- 2 Editor — type your code here
- 3 ▶ Run button
- 4 TERMINAL — output & input
- 5 Status bar — Python version

Step 5 — Create hello.py

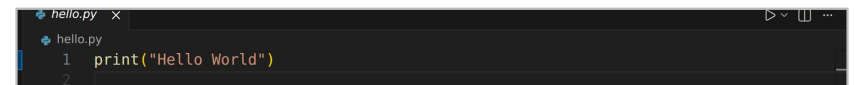
- ▶ In EXPLORER, move the mouse over **COMP1117** → click the **New File...** icon (page with a +)
- ▶ Type **hello.py** and press Enter
- ▶ The file name MUST end with **.py**
 - ▶ the icon turns into the blue-yellow Python logo
 - ▶ the bottom-right corner shows your Python version, e.g. 3.14.4
 - ▶ → VS Code has recognised this as Python and found the translator



bottom-right of the window

Type your first program

```
print("Hello World")
```



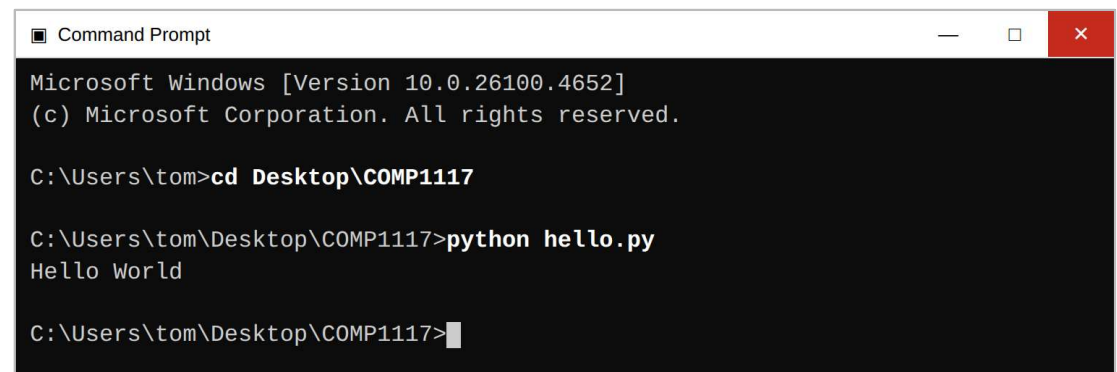
hello.py in the editor — coloured text = VS Code recognised Python

- ▶ Type exactly this line in the editor (the big area on the right)
- ▶ Colours appear as you type: VS Code knows this is Python
- ▶ Use the ENGLISH quote " — NOT the Chinese quotes “ ”
 - ▶ switch your input method to English before typing code
- ▶ Parentheses () and quotes come in pairs — VS Code adds the closing one for you

Step 6 — Run it, way 1: in a terminal (Windows)

A terminal is a window where you type a command, press Enter, and the computer replies in text. Python can run your file from here — VS Code does not even need to be open.

- ▶ ① Press the **Windows key**, type **cmd**, press Enter → a black window “Command Prompt” opens
- ▶ ② Type **cd Desktop\COMP1117** and press Enter (cd = “change directory”: go into that folder)
- ▶ ③ Type **python hello.py** and press Enter
- ▶ ④ The computer replies **Hello World**



```
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Users\tom>cd Desktop\COMP1117

C:\Users\tom\Desktop\COMP1117>python hello.py
Hello World

C:\Users\tom\Desktop\COMP1117>
```

bold = what YOU type. Everything else is the computer talking.

“No such file or directory” / “cannot find the path”? → the terminal is not inside COMP1117. Next slide shows a way that always works.

Step 6 — Run it, way 1: in a terminal (Mac)

A terminal is a window where you type a command, press Enter, and the computer replies in text. Python can run your file from here — VS Code does not even need to be open.

- ▶ ① Press **Cmd + Space**, type **Terminal**, press Enter → a window “Terminal” opens
- ▶ ② Type **cd Desktop/COMP1117** and press Enter (cd = “change directory”: go into that folder)
- ▶ ③ Type **python3 hello.py** and press Enter — on Mac it is python3, with a 3
- ▶ ④ The computer replies **Hello World**



```
tom — zsh — 80x24

Last login: Mon 09:12:41 on ttys000
tom@MacBook-Pro ~ % cd Desktop/COMP1117
tom@MacBook-Pro COMP1117 % python3 hello.py
Hello World
tom@MacBook-Pro COMP1117 % █
```

bold = what YOU type. Everything else is the computer talking.

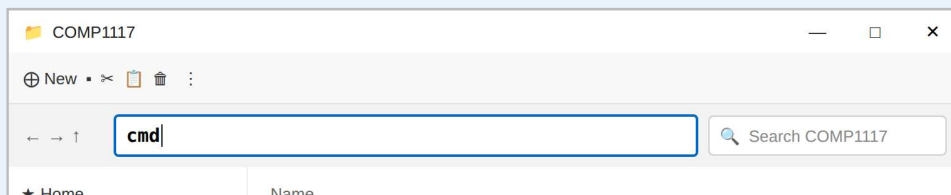
“No such file or directory” / “cannot find the path”? → the terminal is not inside COMP1117. Next slide shows a way that always works.

Can't find the folder?

`cd Desktop\COMP1117` fails if your Desktop lives inside OneDrive, or the folder is somewhere else. Instead, let the computer type the path — start from the folder itself:

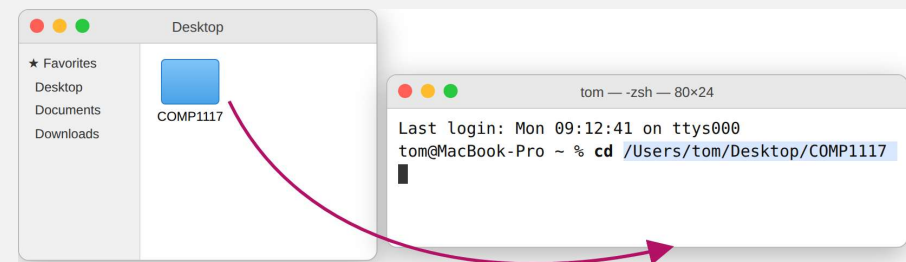
Windows

- ▶ Open the **COMP1117** folder in File Explorer (double-click it on the Desktop)
- ▶ Click in the **address bar** at the top, type `cmd` and press Enter
- ▶ A Command Prompt opens **already inside the folder** — skip the `cd` step



Mac

- ▶ In Terminal type `cd` and one space — do NOT press Enter yet
- ▶ Drag the **COMP1117** folder from the Desktop / Finder into the Terminal window — the full path appears by itself
- ▶ Now press Enter. You are **inside the folder**



Check you are in the right place: Windows `dir` / Mac `ls` → the list must contain `hello.py`. Then run `python hello.py` / `python3 hello.py`.

The same terminal lives inside VS Code

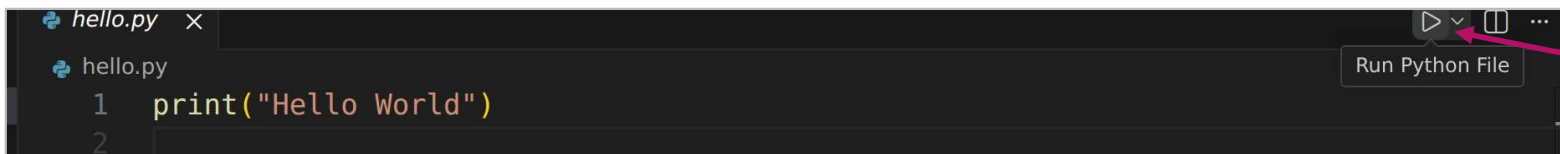
- ▶ Menu **Terminal** → **New Terminal** — a terminal panel opens at the bottom of VS Code
- ▶ It already starts **inside the folder you opened in VS Code**, so you can skip the `cd` step
- ▶ Type `python hello.py` (Mac: `python3 hello.py`) and press Enter
- ▶ Checkpoint: everyone sees **Hello World** before we go on

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  
  
tom@MacBook-Pro COMP1117 % python3 --version  
Python 3.14.4  
tom@MacBook-Pro COMP1117 % python3 hello.py  
Hello World  
tom@MacBook-Pro COMP1117 %
```

same window, same commands —
just docked inside VS Code

Run it, way 2: the ▶ button in VS Code

- ▶ Click the ▶ **Run Python File** button at the TOP-RIGHT of the editor
- ▶ A panel called **TERMINAL** opens at the bottom
- ▶ You should see **Hello World**
- ▶ Same result as way 1 — but with one click. This is how you will normally run your programs.

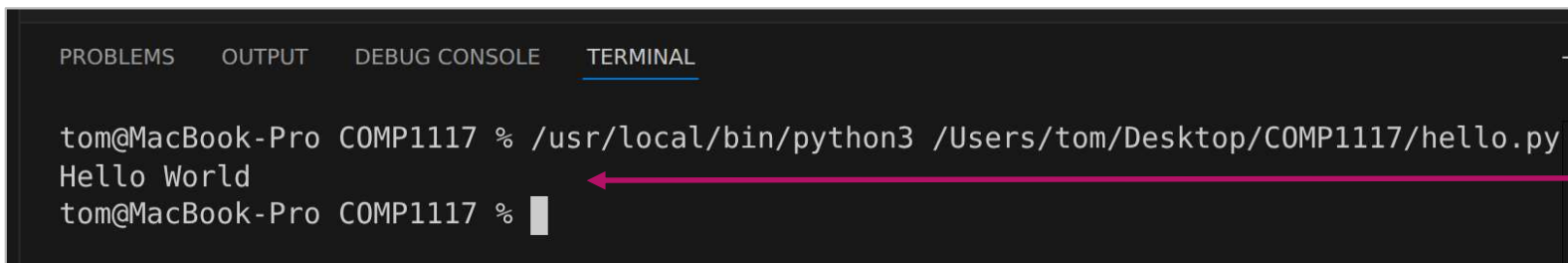


A screenshot of the Visual Studio Code editor interface. The editor window shows a file named 'hello.py' with the following code:

```
1 print("Hello World")
2
```

In the top right corner of the editor, there is a toolbar with several icons. A red arrow points to the 'Run Python File' button, which is represented by a play icon (▶) followed by a checkmark icon.

① click here



A screenshot of the Visual Studio Code terminal panel. The terminal is titled 'tom@MacBook-Pro COMP1117 %'. It shows the command being executed:

```
tom@MacBook-Pro COMP1117 % /usr/local/bin/python3 /Users/tom/Desktop/COMP1117/hello.py
```

The output of the command is displayed on the next line:

```
Hello World
```

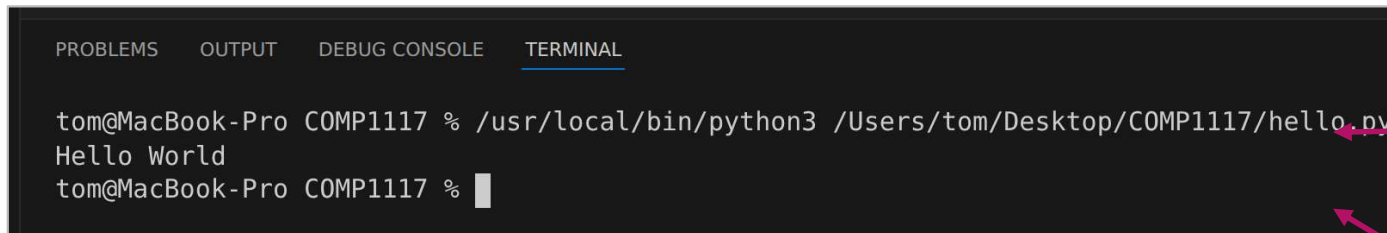
The prompt is then shown again:

```
tom@MacBook-Pro COMP1117 %
```

A red arrow points from the 'output appears here' label to the 'Hello World' output line.

② output appears here

Why does the ▶ button work?



The screenshot shows a VS Code terminal window with the 'TERMINAL' tab selected. The terminal displays the command `/usr/local/bin/python3 /Users/tom/Desktop/COMP1117/hello.py` and its output `Hello World`. A red arrow points from the text 'the command VS Code typed for you' to the command line, and another red arrow points from 'the output of your program' to the output line.

```
tom@MacBook-Pro COMP1117 % /usr/local/bin/python3 /Users/tom/Desktop/COMP1117/hello.py
Hello World
tom@MacBook-Pro COMP1117 %
```

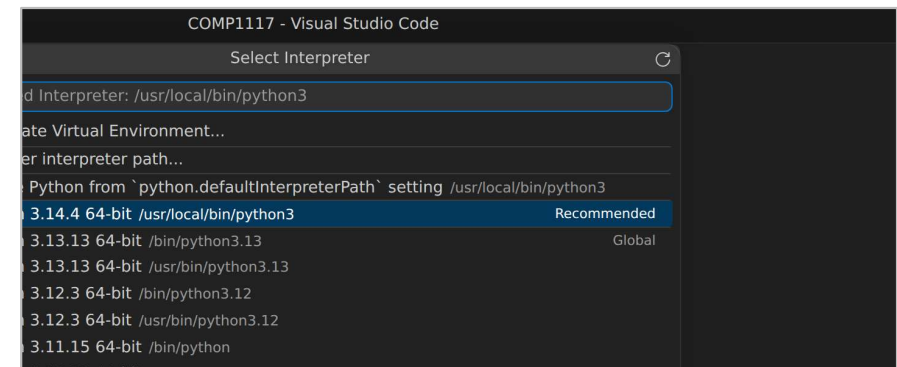
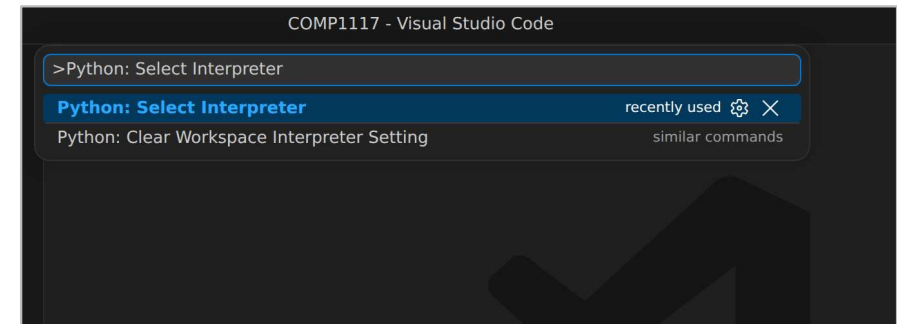
the command VS Code typed for you

the output of your program

- ▶ The ▶ button simply **typed a command** in the terminal: (where Python is) (your file)
- ▶ = “Hey translator (python), please read and execute this file (hello.py)”
- ▶ On Windows the first part looks like `C:\Users\...\python.exe`; on Mac like `/usr/local/bin/python3`. Both are just “where Python is installed”.
- ▶ So VS Code (the notebook) asks Python (the translator) to do the work. **Now you know why you needed both.**

First time only: “Select Interpreter”

- ▶ If the bottom-right corner shows no Python version, or clicking  pops up a list:
- ▶ Press Ctrl+Shift+P (Mac: Cmd+Shift+P) to open the Command Palette
- ▶ Type **Python: Select Interpreter** and press Enter
- ▶ Choose the entry with **3.14** marked **Recommended**
- ▶ You normally do this only once



Programs that ask for input

```
name = input("What is your name? ")  
print("Hello", name)
```

- ▶ Create `ask.py` with these two lines and click ▶
- ▶ The terminal shows `What is your name?` and then **WAITS**
- ▶ **Click INSIDE the terminal panel**, type your name, press Enter
- ▶ Typing in the editor does nothing here — the program is listening in the **terminal**

```
tom@MacBook-Pro COMP1117 % ^C  
tom@MacBook-Pro COMP1117 % /usr/local/bin/python3 /Users/tom/Desktop/COMP1117/ask.py  
What is your name? 
```

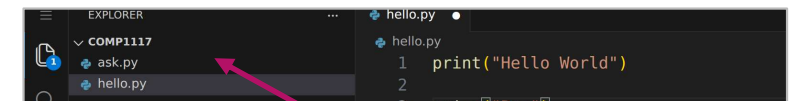
waiting for you — click here and type

```
tom@MacBook-Pro COMP1117 % ^C  
tom@MacBook-Pro COMP1117 % /usr/local/bin/python3 /Users/tom/Desktop/COMP1117/ask.py  
What is your name? Tom  
Hello Tom  
tom@MacBook-Pro COMP1117 % 
```

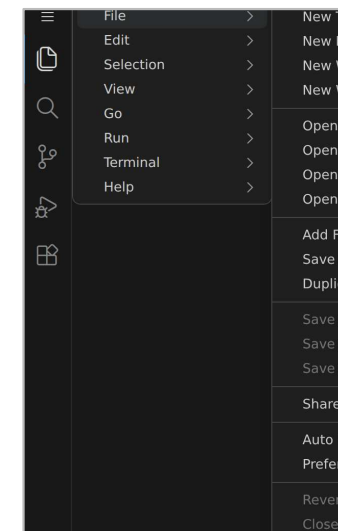
after typing Tom and Enter

Save your work

- ▶ A white dot ● next to the file name means: **changes not saved**
- ▶ Save with **Ctrl+S** (Mac: **Cmd+S**) — the dot becomes x
- ▶ Python runs the file **on the disk**, not what you see on screen. Unsaved = old version runs.
- ▶ Recommended, once: menu **File → Auto Save** — then you never worry about it again



● = not saved



Auto Save

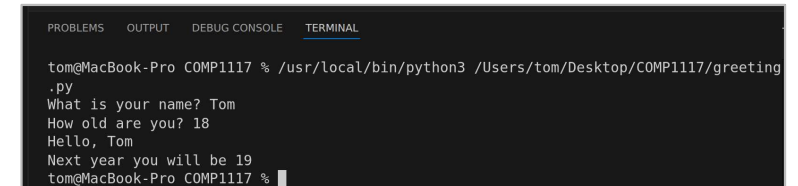
Common problems checklist

You see...	Cause	Fix
Output does not change	File not saved	Ctrl+S / Cmd+S, or turn on Auto Save
can't open file '...hello.py': No such file or directory	Terminal is in a different folder	Use the ► button, or make sure VS Code opened the folder that contains the file
SyntaxError: invalid character '“'	Chinese quotes / brackets from the input method	Switch to English input, retype " and ()
Terminal shows nothing and waits	Program is waiting for input(), or you are inside >>>	Click in the terminal and type; or type exit()
Program never stops	Infinite loop	Click in the terminal, press Ctrl+C
Windows opens the Microsoft Store	PATH not ticked when installing Python	Re-run installer → Modify → tick PATH → restart

Task 1 — Your first program in VS Code

- ▶ 1. In VS Code, with your folder open, create a new file **greeting.py**
- ▶ 2. Type this program (or make up your own questions):

```
name = input("What is your name? ")
age = int(input("How old are you? "))
print("Hello,", name)
print("Next year you will be", age + 1)
```



```
tom@MacBook-Pro COMP1117 % /usr/local/bin/python3 /Users/tom/Desktop/COMP1117/greeting
.py
What is your name? Tom
How old are you? 18
Hello, Tom
Next year you will be 19
tom@MacBook-Pro COMP1117 %
```

what it looks like in the terminal

- ▶ 3. Click ▶, then answer the two questions **in the terminal**
- ▶ 4. Check the output. For Tom and 18 you should see:

```
Hello, Tom
Next year you will be 19
```

Task 1 (continued) — Where is my file?

- ▶ 5. Find the file on your computer: in EXPLORER, **right-click** `greeting.py` →

Windows

- ▶ **Reveal in File Explorer**
- ▶ A File Explorer window opens showing your folder with `greeting.py` selected

Mac

- ▶ **Reveal in Finder**
- ▶ A Finder window opens showing your folder with `greeting.py` selected

- ▶ 6. This is a real file on your computer — a plain `.py` text file in your folder. You could copy it to a USB stick or another computer and Python would run it there too.
- ▶ 7. Run it once more WITHOUT the ▶ button: open the terminal (**Terminal** → **New Terminal**) and type

```
python greeting.py
```

Windows

```
python3 greeting.py
```

Mac

Task 2 — Temperature converter

- 1. Create a new file **temperature.py** and type this program. Complete line 2 yourself using the formula $^{\circ}\text{F} = ^{\circ}\text{C} \times 9 / 5 + 32$

```
celsius = int(input("Temperature in Celsius? "))  
fahrenheit = ...      # your formula here  
print("That is", fahrenheit, "degrees Fahrenheit")
```

Hints

- `input()` gives text — `int()` turns it into a number (line 1 already does this)
- Use `*` for multiply and `/` for divide
- Getting 23.0 or 193.3? Check the brackets and the numbers in your formula
- Stuck? Check the “Common problems checklist” slide, or ask

2. Click ► and type 25 in the terminal. You should see **That is 77.0 degrees Fahrenheit**
3. Test with values you already know the answer to: 0 → 32.0 100 → 212.0
4. Finished? Write `fahrenheit.py` that converts the other way: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5 / 9$

Q & A

- ▶ Please feel free to post on Moodle Forum if you have questions.
- ▶ But don't copy and paste your own code to the Moodle forum.
- ▶ When asking about an error, **paste the error message** (the last few lines in the terminal) or a screenshot of it — it helps us help you.
- ▶ Before you leave today, make sure: Python installed ✓ VS Code + Python extension installed ✓ hello.py runs ✓